

Driven Critical Dynamics in the Tricritical Point

Ting-Long Wang¹, Yi-Fan Jiang^{1*}, and Shuai Yin^{2,3*}¹*School of Physical Science and Technology, ShanghaiTech University, Shanghai 201210, China*²*Guangdong Provincial Key Laboratory of Magnetoelectric Physics and Devices, School of Physics, Sun Yat-Sen University, Guangzhou 510275, China*³*School of Physics, Sun Yat-Sen University, Guangzhou 510275, China*

(Received 24 July 2025; accepted manuscript online 15 September 2025)

The conventional Kibble–Zurek mechanism, describing driven dynamics across critical points based on the adiabatic-impulse scenario (AIS), has attracted broad attention. However, the driven dynamics at the tricritical point with two independent relevant directions have not been adequately studied. Here, we employ the time-dependent variational principle to study the driven critical dynamics at a one-dimensional supersymmetric Ising tricritical point. For the relevant direction along the Ising critical line, the AIS apparently breaks down. Nevertheless, we find that the critical dynamics can still be described by finite-time scaling in which the driving rate has a dimension of $r_\mu = z + 1/\nu_\mu$ with z and ν_μ being the dynamic exponent and correlation length exponent in this direction, respectively. For driven dynamics along another direction, the driving rate has a dimension of $r_p = z + 1/\nu_p$ with ν_p being another correlation length exponent. Our work brings a new fundamental perspective into nonequilibrium critical dynamics near the tricritical point, which could be realized in programmable quantum processors in Rydberg atomic systems.

DOI: [10.1088/0256-307X/42/11/110001](https://doi.org/10.1088/0256-307X/42/11/110001)CSTR: [32039.14.0256-307X.42.11.110001](https://cstr.cn/32039.14.0256-307X.42.11.110001)

1. Introduction. Nonequilibrium quantum critical dynamics represents a significant facet of quantum critical phenomena, attracting extensive investigations from both theoretical and experimental aspects. For the driven critical dynamics, the celebrated Kibble–Zurek (KZ) mechanism gives a unified description of the scaling of topological defects generated after the quench.^[1,2] Crucial to the original KZ mechanism is the adiabatic-impulse scenario (AIS).^[3,4] By tuning a parameter h with dimension $1/\nu$ to cross its critical value h_c linearly with rate R , the evolution is separated into different stages. In the initial stage, the system evolves adiabatically as the correlation time τ is smaller than the transition time $t \sim |h - h_c|/R$. Usually, an energy gap Δ is required to ensure adiabaticity as $\tau \sim \Delta^{-1}$. The adiabaticity ceases to hold inevitably near the critical point since Δ becomes small in the critical region and an impulse region then appears when $\tau > t$.^[1–4] The boundary between these two regions dictates a frozen time.^[3,4] The finite-time scaling (FTS) theory shows that this frozen time scale determines the typical time scale in the impulse region and controls dynamic scaling behaviors of macroscopic quantities.^[5–9] Both the KZ mechanism and FTS have inspired a plethora of investigations.^[10–30] In particular, recently they have been used in rapidly developing quantum devices to prepare quantum states and probe quantum criticality.^[31–39]

On the other hand, as the watershed of second- and first-order phase transitions, tricritical points can display peculiar critical phenomena that have drawn extensive interest in both statistical physics and condensed-

matter physics. For instance, the famous deconfined quantum criticality was recently found to be a tricritical point beyond the Landau–Ginzburg–Wilson paradigm.^[40–44] In Dirac systems, the chiral tricritical point exhibits unusual critical properties beyond quantum-classical mapping.^[45,46] A typical case is the tricritical point in the Ising model.^[47–51] Remarkably, it was shown that emergent space-time supersymmetry, characterizing the invariance of the effective action under exchanging fermion and boson degrees of freedom, can appear therein. Moreover, as its generalization, quantum critical/tricritical points with emergent supersymmetry have been uncovered in various systems.^[51–66]

These intriguing equilibrium behaviors imply that tricritical points can exhibit more diverse nonequilibrium dynamic behaviors. However, previous studies on the dynamics near quantum multicritical points focused on a few integrable models.^[67–70] Thus, the study of driven dynamics associated with tricritical points has yet to reach maturity, leaving substantial room for further exploration. In addition, it was recently proposed that the Ising tricritical point with emergent space-time supersymmetry can be realized in Rydberg atomic systems,^[71] wherein criticality can be probed by external driving.^[31] The requirements from both theoretical and experimental aspects motivate us to explore the driven dynamics across the tricritical point.

In contrast to usual critical points, which possess only one relevant direction, the tricritical point has two independent relevant directions. In particular, one of these

*Corresponding authors. Email: jiangyf2@shanghaitech.edu.cn; yinsh6@mail.sysu.edu.cn

© 2025 Chinese Physical Society and IOP Publishing Ltd. All rights, including for text and data mining, AI training, and similar technologies, are reserved.

directions lies along the critical line. For driving along this direction to cross the tricritical point, the AIS breaks down as a result of the absence of an energy gap in the initial stage.^[18–20,27] Some questions arise as follows: How can we describe the driven dynamics in different relevant directions? Is KZ scaling still applicable along the Ising critical direction?

To answer these questions, in this paper, we explore the driven critical dynamics near the supersymmetric Ising tricritical point by changing both two relevant parameters. To change the parameter along the Ising critical phase with gapless Majorana fermionic excitations to cross the tricritical point, the dynamic scaling properties of both the bosonic order parameter and the fermion correlation are explored. We shown that the driving rate has a dimension of $r_\mu = z + 1/\nu_\mu$, where z is the dynamic exponent and ν_μ is the correlation length exponent of the tricritical point in this direction. Accordingly, we confirm that KZ scaling is still applicable despite the breakdown of AIS. We attribute the validity of KZ scaling to satisfying the condition $z' < z + 1/\nu_\mu$, where z' is the dynamic exponent of the gapless phase.^[27] In addition, for comparison, we also explore driven dynamics along other directions and confirm that KZ scaling, characterized by another rate exponent $r_p = z + 1/\nu_p$, works. Possible experimental realizations in programmable quantum processors in Rydberg atomic systems are discussed.

2. *Model and Static Properties.* The Hamiltonian H of the model we considered is^[51]

$$H = H_\sigma + H_\mu + H_{\sigma\mu} \quad (1)$$

with

$$H_\sigma = J_\sigma \sum_i \sigma_i^z \sigma_{i+1}^z - h_\sigma \sum_i \sigma_i^x, \quad (2a)$$

$$H_\mu = J_\mu \sum_i (\mu_{i,a}^z \mu_{i,b}^z + \mu_{i,b}^z \mu_{i+1,a}^z) - h_\mu \sum_i (\mu_{i,a}^x + \mu_{i,b}^x), \quad (2b)$$

$$H_{\sigma\mu} = g \sum_i (\sigma_i^x \mu_{i,a}^z - \sigma_i^z \mu_{i+1,b}^z), \quad (2c)$$

in which $\sigma^{x,z}$ and $\mu^{x,z}$ are Pauli matrices in the x and z directions, J_σ and J_μ are the coupling coefficients, h_σ and h_μ are the strengths of the transverse fields, and g is the coupling between two spin chains. In the following, we choose J_σ and J_μ as unity. After applying the Jordan–Wigner transformation on the σ -spin, the Hamiltonian H can be converted to^[51]

$$H = -i \sum_j [1 - g \mu_{j+1/2}^z + (-1)^j (h_\sigma - 1)] \chi_j \chi_{j+1} + \sum_j \mu_{j-1/2}^z \mu_{j+1/2}^z - h_\mu \sum_j \mu_{j+1/2}^x, \quad (3)$$

in which χ represents the Majorana fermion operator.

It was shown that a tricritical point appears at $h_{\sigma,c} = 1$ and $h_{\mu,c} \approx 1.27$ for $g = 0.5$ [Fig.1, details are in the Supplementary Materials (SM)]. At the tricritical point, supersymmetry emerges, featured by identical anomalous

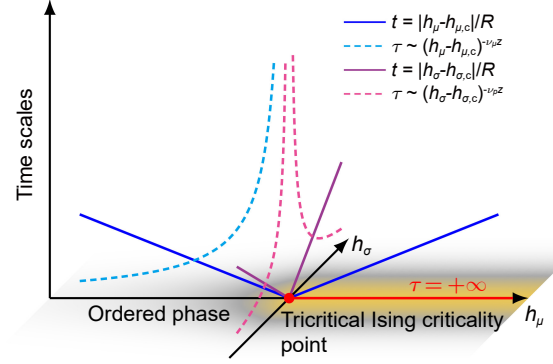


Fig. 1. Sketch of the phase diagram near the tricritical point of the Ising model and the protocol for driven dynamics. Around the tricritical dimension, there are two independent relevant directions. One is the h_μ -direction for $h_\sigma = 1$. Although the conventional AIS breaks down for driven dynamics from the Ising critical phase along this direction, we show that the dynamic scaling behaviors satisfy the FTS form in which the driving rate R has the dimension of $r_\mu = z + 1/\nu_\mu$. The driven dynamics along other relevant direction is explored by tuning h_σ to cross the tricritical point, which exhibits a distinct FTS form in which R has the dimension of $r_p = z + 1/\nu_p$.

dimensions for both the bosonic order parameter field μ^z and the fermionic operator χ , namely $\eta_b = \eta_f = 2/5$.^[51]

There are two independent relevant directions around the tricritical point. One is the h_μ -direction with fixed $h_\sigma = 1$. Along this direction, for large h_μ , the order parameter, namely the expectation value of μ^z , is zero. In this way, the system is in the Ising critical phase and the fermion excitations are gapless. For small h_μ , $\langle \mu^z \rangle$ becomes finite due to spontaneous symmetry breaking. Accordingly, the Majorana fermion gains a mass via the term $g\mu^z$. The correlation length exponent in the h_μ -direction is $\nu_\mu = 5/4$. In the perpendicular direction, namely the p -direction, the correlation length exponent is $\nu_p = 5/9$. In practice, determining this direction exactly is very subtle.^[72] However, it was shown that for any arbitrary direction that has a nonzero component in the p -direction, the critical behavior is dominated by the variable in the p -direction since $1/\nu_p > 1/\nu_\mu$.^[73]

In the following, we utilize the time-dependent variational principle (TDVP)^[74] with a matrix product state (MPS) ansatz to simulate the driven dynamics. The bond dimension of MPS is taken up to $D = 1000$ and time step $\tau = 0.02$ – 0.04 are taken to ensure the accuracy of the TDVP simulation. Although the original KZ mechanism concentrates on the scaling of the dependence of topological defects on the driving rate,^[1–4] FTS generalizes this scaling theory to more general quantities in the whole critical region. In this paper, we focus on dynamic scaling behaviors exactly at the tricritical point for driven dynamics along different directions.

3. *Driven Dynamics along h_μ -Direction.* We begin with the driven dynamics by fixing $h_\sigma = 1$ and decreasing h_μ as $h_\mu = h_{\mu 0} - Rt$ with $h_{\mu 0} = 1.8$. The corresponding initial state is in the Ising critical phase that hosts gapless Majorana excitations. Therefore, the correlation time in

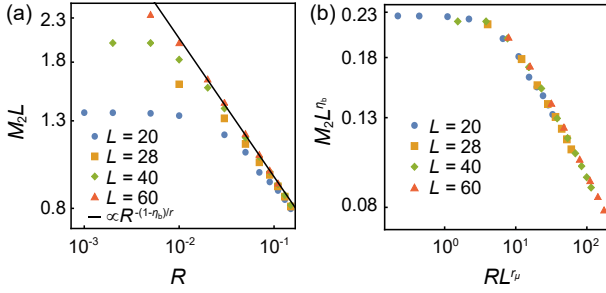


Fig. 2. Scaling behavior of the order parameter M_2 at the tricritical point when the system is driven from the gapless Ising critical phase by decreasing h_μ . The solid line in (a) shows the power law $M_2 \propto R^{-(1-\eta_b)/r_\mu}$ fitted to the data for $L = 60$ in the large R region. The rescaled curves in (b) collapse onto each other.

the initial stage is divergent and always larger than the transition time, implying the breakdown of the AIS in the original KZ dynamics.

We first study the dynamics of the square of the order parameter defined as $M_2 \equiv (1/L) \sum_i \langle \mu_0^z \mu_i^z \rangle$ for large R . Figure 2 shows the dependence of M_2 on R at the tricritical point. From Fig. 2(a), we can see that $M_2 \propto L^{-1}$ for a fixed R . Moreover, by comparing with a curve of the power function with exponent $(1 - \eta_b)/r_\mu$, in which $\eta_b = 2/5$ and $r_\mu = z + 1/\nu_\mu$, we find the scaling relation of M_2 as

$$M_2(R, L) \propto L^{-1} R^{-(1-\eta_b)/r_\mu}. \quad (4)$$

Physically, equation (4) reflects the memory of the initial state in the universal dynamics of the tricritical point under external driving. First, $M_2 \propto L^{-1}$ stems from the property of M_2 for the initial state, since the μ^z -spin in H_μ -chain is short-range correlated in the large h_μ region. Second, the critical exponents of the tricritical point enter Eq. (4) via the driving rate R . Since M_2 has a dimension of η_b , equation (4) demonstrates that R has a dimension of r_μ , which is consistent with KZ scaling.

Accordingly, for a general R , the dynamic scaling of M_2 should obey

$$M_2(R, L) = L^{-1} R^{-(1-\eta_b)/r_\mu} \mathcal{F}(RL^{r_u}), \quad (5)$$

in which $\mathcal{F}$ is a scaling function. Note that Eq. (5) is just the FTS form in the usual critical point for driven dynamics from the disordered phase.^[8,24] In addition, equation (5) can be converted to $M_2(R, L) = L^{-\eta_b} \mathcal{F}_1(RL^{r_u})$, from which the usual finite-size scaling of $M_2(R, L) \propto L^{-\eta_b}$ at the tricritical point can be recovered.^[58] After rescaling M_2 and R as $M_2 L^{\eta_b}$ and RL^{r_u} , respectively, we find that the rescaled curves collapse onto each other as shown in Fig. 2, confirming the scaling behavior in Eq. (5). These results demonstrate that the KZ scaling still holds albeit the breakdown of AIS for gapless initial state.

To further validate this conclusion, we then study the dynamic scaling of the fermion correlation defined as $C_f \equiv 1/L \sum_i \langle \chi_i \chi_{i+L/2} \rangle$ at the tricritical point. From Fig. 3(a), we find that for large R , the scaling relation of C_f on R is

$$C_f(R, L) \propto L^{-1} R^{\eta_f/r_\mu} \quad (6)$$

(see SM for details). Similarly, equation (6) also combines the initial information, which is reflected in $C_f \propto L^{-1}$ stemming from the scaling of gapless Majorana excitations, with tricritical properties manifested in the exponent of R . Moreover, since C_f has the dimension of $1 + \eta_f$, equation (6) demonstrates that R has a dimension of r_μ , confirming the validity of KZ scaling. Accordingly, the FTS form of C_f should satisfy

$$C_f(R, L) = L^{-1} R^{\eta_f/r_\mu} \mathcal{F}_2(RL^{r_u}), \quad (7)$$

which can be modified to $C_f(R, L) = L^{-1-\eta_f} \mathcal{F}_3(RL^{r_u})$, restoring the finite-size scaling $C_f(R, L) \propto L^{-1-\eta_f}$ for small R .^[58] This FTS form is confirmed in the scaling collapse of the rescaled curves of $C_f L^{1+\eta_f}$ versus RL^{r_u} shown in Fig. 3(b).

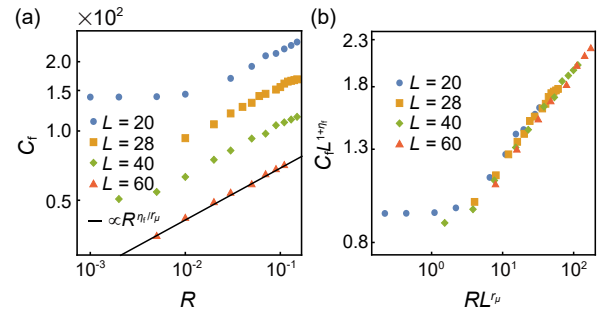


Fig. 3. Scaling behavior of the half-chain fermion correlation C_f at the tricritical point when the system is driven from the gapless Ising critical phase by decreasing h_μ . The solid line in (a) shows the power law $C_f \propto R^{\eta_f/r_\mu}$ fitted to the data for $L = 60$ in the large R region. The rescaled curves in (b) collapse onto each other.

In the equilibrium study of the tricritical point, the scaling behavior of the entanglement entropy is an important indicator to diagnose the critical properties. In particular, the half-chain entanglement entropy S satisfies $S = (c/3) \log L$ in the critical phase/point with c being the central charge. In the Ising critical phase, $c = c_I = 1/2$, while at the tricritical point, $c = c_T = 7/10$.^[51] For KZ dynamics with AIS in usual critical point, scaling of entanglement entropy was also studied.^[75] Incorporating the scaling dimension of R derived above, one can obtain the FTS form of S at the tricritical point for driven dynamics from the Ising critical phase as follows:

$$S(R, L) = \frac{c_T}{3} \log L + \mathcal{F}_4(RL^{r_u}). \quad (8)$$

For large R , S should retain the memory of the initial Ising critical phase. This requires the scaling function $\mathcal{F}_4$ to obey $\mathcal{F}_4(RL^{r_u}) = [(c_I - c_T)/3r_u] \log(RL^{r_u})$ such that Eq. (8) tends to

$$S(R, L) = \frac{c_I}{3} \log L + \frac{(c_I - c_T)}{3r_u} \log R. \quad (9)$$

Figure 4(a) shows the curves of $S - (c_I/3) \log L$ versus $\log R$. In the large R region, the curve for large L can be described by a line with the slope close to $(c_I - c_T)/3r_u$, consistent with Eq. (9). The FTS form of S in Eq. (8) is confirmed by the collapse of the rescaled curves of $S - (c_T/3) \log L$ versus RL^{r_u} illustrated in Fig. 4(b). Note

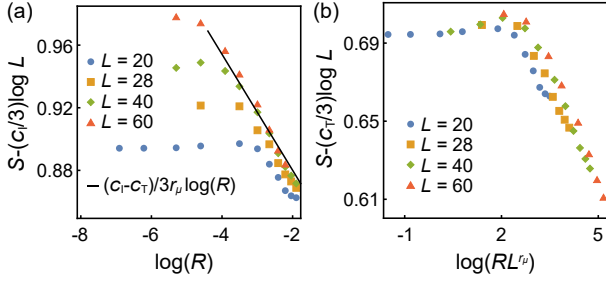


Fig. 4. Scaling behavior of the half-chain entanglement entropy S at the tricritical point when the system is driven from the gapless Ising critical phase by decreasing h_μ . The solid line in (a) shows the logarithmic law $S - (c_I - c_T)/3r_\mu \propto [(c_I - c_T)/3r_\mu] \log R$ fitted to the data for $L = 60$ in the large R region. The rescaled curves in (b) collapse onto each other.

that the coefficient before $\log R$ incorporates both the central charges of Ising criticality and Ising tricriticality. This scaling relation is remarkably distinct from the entanglement dynamics in usual critical points, wherein only the central charge of the critical point appears in the coefficient of $\log R$.^[75]

The original KZ mechanism is based on the AIS, which requires the existence of an energy gap to protect the adiabaticity in the initial stage. In contrast, the above results reveal that for driven dynamics along the critical line, wherein the system is in a gapless state, although the AIS breaks down, the KZ scaling remains valid. To explore this reason, we inspect Eqs. (2) and (3) with $h_\sigma = 1$. When the system is driven from critical phase $h_\mu > h_{\mu,c}$, $\langle \mu^z \rangle$ remains zero in the initial stage. Accordingly, at mean-field level, the coupling between two Ising chains in Eq. (2), i.e., Eq. (2c), vanishes. Thus, the eigenstates of Eq. (2a) or the gapless Majorana fermion part in Eq. (3) remain intact. Consequently, the excitations in the initial stage are absent. In fact, it was shown that if the contributions beyond the mean field are taken into account, only the velocity of the Majorana fermion changes, while the linear dissipation of the gapless Majorana fermion keeps invariant.^[76] In this case, the density of excitation in the gapless phase obeys $n_{\text{ex}} \propto R^{1/z'}$ with $z' = 1$ being the dynamic exponent of the Ising critical phase.^[77] In contrast, the KZ scaling of n_{ex} for driving across the tricritical point is $n_{\text{ex}} \propto R^{1/r_\mu}$.^[10–12] Since $1/z' > 1/r_\mu$, for KZ dynamics with $R < 1$, critical properties near the tricritical point rather than those in the critical phase dominate. Note that recently a similar criterion in usual critical points for validating KZ scaling with a gapless initial state was proposed,^[27] which gives a unified explanation for critical-point-dominated scaling behaviors reported previously.^[18–20] Here, we successfully generalize this criterion to include cases at and near the tricritical point.

As a contrast, we then explore the driven dynamics by fixing $h_\sigma = 1$ and increasing h_μ as $h_\mu = h_{\mu,0} + Rt$ from the initial state $h_{\mu,0} = 0.6$ in the ordered phase with gapped fermion excitation. In this case, the dynamical process obeys the AIS. Thus, The driven dynamics near the tricritical point is similar to that in usual critical points with

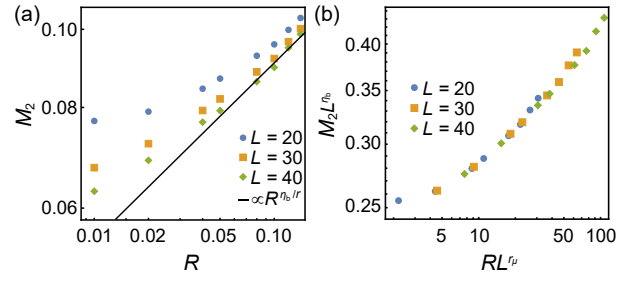


Fig. 5. Scaling behavior of M_2 at the tricritical point for increasing h_μ . The solid line in (a) shows the logarithmic law $M_2 \propto R^{\eta_b/r_\mu}$ fitted to the data for $L = 40$ in the large R region. The rescaled curves in (b) collapse onto each other.

an ordered initial state. Accordingly, the order parameter M_2 at the tricritical point should conform to^[5,8]

$$M_2(R, L) = R^{\eta_b/r_\mu} \mathcal{F}_5(RL^{r_\mu}), \quad (10)$$

where the memory from the ordered initial state, in which M_2 is almost independent of L , makes $M_2(R, L) \propto R^{\eta_b/r_\mu}$ for large R .^[5,8,27] This is verified in Fig. 5(a). In addition, figure 5(b) shows a nice collapse of the rescaled curves as expected according to Eq. (10).

4. Driven Dynamics along h_σ -Direction. For driven dynamics along any path other than the h_μ -direction to cross the tricritical point, the AIS should be followed since the initial state has energy gap which can ensure the adiabaticity in the initial stage. A remarkable point is that, different from usual critical points, the tricritical point possesses two independent relevant directions. Although the definite one with $\nu_\mu = 5/4$ is along the critical line, determining its perpendicular counterpart p -direction with $\nu_p = 5/9$ seems quite subtle.^[72] For driven dynamics along an arbitrary dimension, which can be decomposed into a component in the h_μ -direction and a component in the p -direction, the driving rate R can consequently have two dimensions, namely $r_\mu = z + 1/\nu_\mu$ and $r_p = z + 1/\nu_p$. Moreover, since $r_p > r_\mu$, the driven dynamics should be dominated by the p -direction.

To verify this inference, we take the driven dynamics along the h_σ -direction with h_μ fixed at $h_{\mu,c}$ as an example. Figure 6 shows the dependence of M_2 on R at the tricritical point for a series of system sizes. For large R , we find

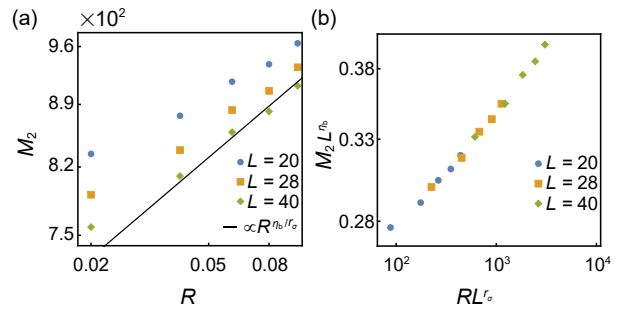


Fig. 6. Scaling behavior of M_2 at the tricritical point for increasing h_σ . The solid line in (a) shows the logarithmic law $M_2 \propto R^{\eta_b/r_\sigma}$ with $r_\sigma = r_p$ fitted to the data for $L = 40$ in the large R region. The rescaled curves in (b) collapse onto each other.

from Fig. 6(a) that

$$M_2(R, L) \propto R^{\eta_b/r_\sigma} \quad (11)$$

with $r_\sigma = r_p$ for large system sizes. Scaling analyses based on the dimension of M_2 suggest that the dimension of R is r_p , confirming the discussion above. Moreover, this conclusion can be further verified by the full FTS form of M_2 ,

$$M_2(R, L) = R^{\eta_b/r_\sigma} \mathcal{F}_6(RL^{r_\sigma}). \quad (12)$$

The scaling collapse of rescaled curves of M_2 versus R for different L shown in Fig. 6(b) confirms Eq. (12), demonstrating that the typical exponent along h_σ -direction is $r_\sigma = r_p$.

5. Discussion. Here we briefly discuss the experimental feasibility of our numerical results. It was recently proposed that this supersymmetric tricritical point can be realized in Rydberg atomic ladder with two species of atoms, denoted as A and B , respectively.^[71] Each atom can be described by two-level pseudo-spin states: the ground state $|g\rangle$ (spin-up) and the Rydberg state $|r\rangle$ (spin-down). A laser can induce coupling between these states and determines the Rabi frequency Ω and detuning Δ through the laser strength and frequency, respectively. The Hamiltonian reads

$$H = \sum_i \left(\frac{\Omega}{2} \tau_i^x - \Delta n_i \right) + \sum_{ij} \frac{C_{XY}}{r_{ij}^6} n_i n_j, \quad (13)$$

in which $\tau^x \equiv |r\rangle\langle g| + |g\rangle\langle r|$, $n_i \equiv (1 + \tau^z)/2$ is the number of Rydberg excitations, X and Y represent the species of the atoms i and j , and r_{ij} is the distance between atoms. Accordingly, equation (13) is quite similar to two interacting Ising chains described in Eq. (2). Moreover, it was shown that to realize the supersymmetric tricritical point, C_{AA} and C_{BB} should be positive, whereas C_{AB} should be negative. By tuning Δ and Ω , the tricritical point can be reached. In Rydberg arrays, changing Δ and Ω to cross the tricritical point can be readily realized.^[78]

In addition, in our paper, we consider the dynamics of the order parameter M and the correlation between the Majorana fermions χ . It was shown that the order parameter M is proportional to $n \equiv n_1 - n_2$ in which n_1 and n_2 denote the number of Rydberg states in two legs of the ladder, and the Majorana fermion correlation $\langle \chi_i \chi_j \rangle \propto (n_{i1} - n_{i2}) \prod_{k=i+1}^{j-1} \text{SWAP}_k(n_{j1} - n_{j2})$, where the SWAP_k operator applies an exchange between two legs of the ladder at site k onto the quantum state.^[71] Although the non-local properties in the correlation can bring some difficulties in experiments, it was argued that the fermion correlation can still be detected with the aid of the quantum analog-to-digital converter.^[71] Accordingly, it is expected that our results can be further examined in these experiments.^[31–39]

6. Summary. In summary, this paper systematically explores the driven critical dynamics at the 1D Ising tricritical point. We find that for driven dynamics along the critical line to cross the tricritical point, the KZ scaling characterized by exponent $r = z + 1/\nu_\mu$ still stands even

though the AIS breaks down in the critical region. We point out that the reason is that the excitations in the initial stage are much fewer compared to those at the critical point. In addition, we have also revealed that the driven dynamics along other direction are described by KZ scaling with a typical exponent $r = z + 1/\nu_p$, due to the dominating perpendicular relevant direction. This work provides a profound new perspective on nonequilibrium critical dynamics near tricritical points. Possible experimental realizations in Rydberg atomic arrays are also discussed.

Note Added. We thank the authors of Ref. [78] for helpful discussion. In Ref. [78], driven dynamics of the tricritical point in another model belonging to the same universality class is considered, and consistent results are obtained.

Acknowledgments. This work was supported by the National Natural Science Foundation of China (Grant Nos. 12222515, 12075324 for S. Yin, and 12347107, 1257-4160 for Y. F. Jiang), the National Key R&D Program of China (Grant No. 2022YFA1402703 for Y. F. Jiang), the Science and Technology Projects in Guangdong Province (Grant No. 2021QN02X561 for S. Yin), and the Science and Technology Projects in Guangzhou City (Grant No. 2025A04J5408 for S. Yin).

References

- [1] Kibble T W B 1976 *J. Phys. A: Math. Gen.* **9** 1387
- [2] Zurek W H 1985 *Nature* **317** 505
- [3] Dziarmaga J 2010 *Adv. Phys.* **59** 1063
- [4] Polkovnikov A, Sengupta K, Silva A, and Vengalattore M 2011 *Rev. Mod. Phys.* **83** 863
- [5] Zhong F and Xu Z 2005 *Phys. Rev. B* **71** 132402
- [6] Gong S, Zhong F, Huang X, and Fan S 2010 *New J. Phys.* **12** 043036
- [7] Yin S, Mai P, and Zhong F 2014 *Phys. Rev. B* **89** 094108
- [8] Huang Y, Yin S, Feng B, and Zhong F 2014 *Phys. Rev. B* **90** 134108
- [9] Feng B, Yin S, and Zhong F 2016 *Phys. Rev. B* **94** 144103
- [10] Zurek W H, Dorner U, and Zoller P 2005 *Phys. Rev. Lett.* **95** 105701
- [11] Dziarmaga J 2005 *Phys. Rev. Lett.* **95** 245701
- [12] Polkovnikov A 2005 *Phys. Rev. B* **72** 161201
- [13] Du K, Fang X, Won C, De C, Huang F T, Xu W, You H, Gómez-Ruiz F J, del Campo A, and Cheong S W 2023 *Nat. Phys.* **19** 1495
- [14] Ko B, Park J W, and Shin Y 2019 *Nat. Phys.* **15** 1227
- [15] Maegochi S, Ienaga K, and Okuma S 2022 *Phys. Rev. Lett.* **129** 227001
- [16] Qiu L Y, Liang H Y, Yang Y B, Yang H X, Tian T, Xu Y, and Duan L M 2020 *Sci. Adv.* **6** eaba7292
- [17] Sunami S, Singh V P, Garrick D, Beregi A, Barker A J, Luksch K, Bentine E, Mathey L, and Foot C J 2023 *Science* **382** 443
- [18] Divakaran U, Dutta A, and Sen D 2008 *Phys. Rev. B* **78** 144301
- [19] Suzuki S and Dutta A 2015 *Phys. Rev. B* **92** 064419
- [20] Deng S, Ortiz G, and Viola L 2009 *EPL* **84** 67008
- [21] Chandran A, Erez A, Gubser S S, and Sondhi S L 2012 *Phys. Rev. B* **86** 064304
- [22] Kolodrubetz M, Clark B K, and Huse D A 2012 *Phys. Rev. Lett.* **109** 015701
- [23] Yin S, Huang G Y, Lo C Y, and Chen P 2017 *Phys. Rev. Lett.* **118** 065701
- [24] Liu C W, Polkovnikov A, and Sandvik A W 2014 *Phys. Rev. B* **89** 054307

- [25] Liu C W, Polkovnikov A, and Sandvik A W 2015 *Phys. Rev. Lett.* **114** 147203
- [26] Clark L W, Feng L, and Chin C 2016 *Science* **354** 606
- [27] Zeng Z, Yu Y K, Li Z X, Li Z X, and Yin S 2025 *Nat. Commun.* **16** 2041
- [28] Zeng Z, Yu Y K, Li Z X, and Yin S 2025 *Phys. Rev. B* **112** L060301
- [29] Deng M, Sun Z, and Li F 2025 *Phys. Rev. Lett.* **134** 010409
- [30] Shu Y R, Jian S K, Sandvik A W, and Yin S 2025 *Nat. Commun.* **16** 3402
- [31] Keesling A, Omran A, Levine H *et al.* 2019 *Nature* **568** 207
- [32] King A D, Raymond J, Lanting T *et al.* 2023 *Nature* **7** 61
- [33] Ebadi S, Wang T T, Levine H, Keesling A, Semeghini G, Omran A, Bluvstein D, Samajdar R, Pichler H, HoWW, Choi S, Sachdev S, Greiner M, Vuletić V, and Lukin M D 2021 *Nature* **595** 227
- [34] Soto Garcia J and Chepiga N 2024 *Phys. Rev. B* **110** 125113
- [35] King A D, Nocera A, Rams M M *et al.* 2024 [arXiv:2403.00910](https://arxiv.org/abs/2403.00910) [quant-ph]
- [36] Dupont M and Moore J E 2022 *Phys. Rev. B* **106** L041109
- [37] Li B W, Wu Y K, Mei Q X, Yao R, Lian W Q, Cai M L, Wang Y, Qi B X, Yao L, He L, Zhou Z C, and Duan L M 2023 *PRX Quantum* **4** 010302
- [38] Ebadi S, Keesling A, Cain M, Wang T T, Levine H, Bluvstein D, Semeghini G, Omran A, Liu J G, Samajdar R, Luo X Z, Nash B, Gao X, Barak B, Farhi E, Sachdev S, Gemelke N, Zhou L, Choi S, Pichler H, Wang S T, Greiner M, Vuletić V, and Lukin M D 2022 *Science* **376** 1209
- [39] Zhang T, Wang H, Zhang W, Wang Y, Du A, Li Z, Wu Y, Li C, Hu J, Zhai H, and Chen W 2025 *Phys. Rev. Lett.* **135** 093403
- [40] Senthil T, Vishwanath A, Balents L, Sachdev S, and Fisher M P A 2004 *Science* **303** 1490
- [41] Zhao B, Takahashi J, and Sandvik A W 2020 *Phys. Rev. Lett.* **125** 257204
- [42] Chester S M and Su N 2024 *Phys. Rev. Lett.* **132** 111601
- [43] Li Z 2022 *J. High Energ. Phys.* **5** (2022).
- [44] Takahashi J, Shao H, Zhao B, Guo W, and Sandvik A W 2024 [arXiv:2405.06607](https://arxiv.org/abs/2405.06607) [cond-mat.str-el]
- [45] Yin S, Jian S K, and Yao H 2018 *Phys. Rev. Lett.* **120** 215702
- [46] Yin S and Zuo Z Y 2020 *Phys. Rev. B* **101** 155136
- [47] Friedan D, Qiu Z, and Shenker S 1984 *Phys. Rev. Lett.* **52** 1575
- [48] Friedan D, Qiu Z, and Shenker S 1985 *Phys. Lett. B* **151** 37
- [49] Zamolodchikov A B 1986 *Sov. J. Nucl. Phys.* **44** 529
- [50] O'Brien E and Fendley P 2018 *Phys. Rev. Lett.* **120** 206403
- [51] Grover T, Sheng D N, and Vishwanath A 2014 *Science* **344** 280
- [52] Jian S K, Jiang Y F, and Yao H 2015 *Phys. Rev. Lett.* **114** 237001
- [53] Jian S K, Lin C H, Maciejko J, and Yao H 2017 *Phys. Rev. Lett.* **118** 166802
- [54] Balents L, Fisher M P A, and Nayak C 1998 *Int. J. Mod. Phys. B* **12** 1033
- [55] Fendley P, Schoutens K, and de Boer J 2003 *Phys. Rev. Lett.* **90** 120402
- [56] Lee S S 2007 *Phys. Rev. B* **76** 075103
- [57] Ponte P and Lee S S 2014 *New J. Phys.* **16** 013044
- [58] Li Z X, Jiang Y F, and Yao H 2017 *Phys. Rev. Lett.* **119** 107202
- [59] Rahmani A, Zhu X, Franz M, and Affleck I 2015 *Phys. Rev. Lett.* **115** 166401
- [60] Yu Y and Yang K 2010 *Phys. Rev. Lett.* **105** 150605
- [61] Huijse L, Bauer B, and Berg E 2015 *Phys. Rev. Lett.* **114** 090404
- [62] Bauer B, Huijse L, Berg E, Troyer M, and Schoutens K 2013 *Phys. Rev. B* **87** 165145
- [63] Li Z X, Vaezi A, Mendl C B, and Yao H 2018 *Sci. Adv.* **4** eaau1463
- [64] Ma K K W, Wang R, and Yang K 2021 *Phys. Rev. Lett.* **126** 206801
- [65] Li C, Lantagne-Hurtubise E, and Franz M 2019 *Phys. Rev. B* **100** 195146
- [66] Yu J, Roiban R, Jian S K, and Liu C X 2019 *Phys. Rev. B* **100** 075153
- [67] Divakaran U, Mukherjee V, Dutta A, and Sen D 2009 *J. Stat. Mech.* **2009** P02007
- [68] Deng S, Ortiz G, and Viola L 2009 *Phys. Rev. B* **80** 241109
- [69] Mukherjee V and Dutta A 2010 *EPL* **92** 37004
- [70] Patra A, Mukherjee V, and Dutta A 2011 *J. Phys.: Conf. Ser.* **297** 012008
- [71] Li C, Liu S, Wang H, Zhang W, Li Z X, Zhai H, and Gu Y 2024 *Phys. Rev. Lett.* **133** 223401
- [72] Sandvik A W 2024 [arXiv:2406.12681](https://arxiv.org/abs/2406.12681) [cond-mat.stat-mech]
- [73] Kwak W, Jeong J, Lee J, and Kim D H 2015 *Phys. Rev. E* **92** 022134
- [74] Haegeman J, Cirac J I, Osborne T J, Pižorn I, Verschelde H, and Verstraete F 2011 *Phys. Rev. Lett.* **107** 070601
- [75] Cincio L, Dziarmaga J, Rams M M, and Zurek W H 2007 *Phys. Rev. A* **75** 052321
- [76] Tang H K, Leaw J N, Rodrigues J N B, Herbut I F, Sen-gupta P, Assaad F F, and Adam S 2018 *Science* **361** 570
- [77] Polkovnikov A and Gritsev V 2008 *Nat. Phys.* **4** 477
- [78] Wang H, Li X, and Li C 2025 [arXiv:2505.12979](https://arxiv.org/abs/2505.12979) [cond-mat.quant-gas]